

controller controls MOSFET switching. The inductor is repeatedly 'charged' and 'discharged' to produce an output voltage higher than the supply voltage.

The Schottky diode blocks and transfers the stored energy to the capacitor and the load. The capacitor acts as a ripple filter.

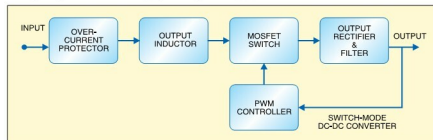


Fig. 4: Block diagram of the vehicle power adaptor

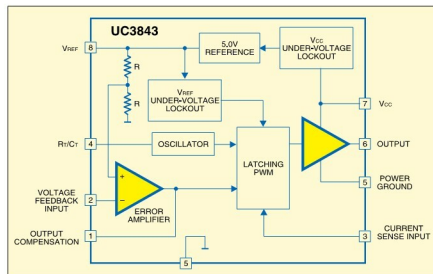


Fig. 5: Block diagram of UC3843 IC

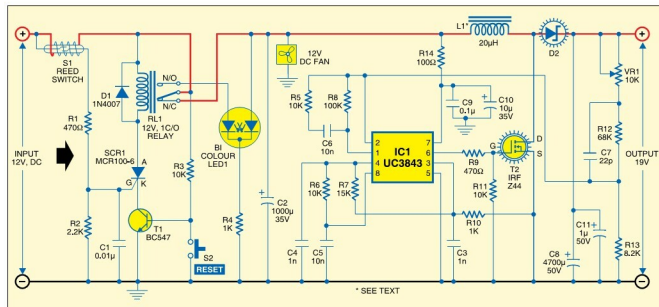


Fig. 6: Circuit of vehicle power adaptor

Fig. 5 shows the block diagram of the UC3843 (IC1), while Fig. 6 shows the circuit of the vehicle power adaptor. All the control functions are performed by IC1. This device produces the PWM switching signal for the power switch. It contains a PWM controller output signal and an internal voltage reference.

The output voltage of the adaptor is divided by resistors R12 and R13 and potentiometer VR1 and fed to the voltage feedback pin (inverting input pin 2) of the error amplifier of IC1. Capacitor C7 improves the stability of the amplifier, as do resistors R5 and R8 and capacitor C6 connected between output pin 1 and inverting input of the differential amplifier.

The clock frequency is determined by the resistor-capacitor (R-C) network formed by resistor R6 and capacitor C4. Capacitor C5 provides decoupling of the internal reference voltage at pin 8. The supply decoupling network

formed by resistor R14 and capacitors C9 and C10 on supply pin 7 is important for reliable op-

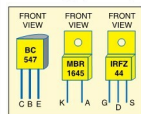


Fig. 7: Pin configurations of BC547, MBR1645 and IRFZ44